Uncertainty and failure analysis

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One complete confidence and review-policy record.

Keep inputs, predictions, counts, missing reviewer outcomes and next checks together.

Keep one habit: Recall → W1–W3/W5 → independent score table → project report → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Transfer learning. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Embeddings and neighbourhoods. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A model can be confidently wrong. Use these supplied **output**s for a fictional image classifier: class 1 means a triangle is present; class 0 means it is absent.

  The values p1 are probability estimates for class 1, invented for this exercise rather than produced by a trained model. Predict 1 when p1 >= 0.5 and 0 otherwise. The selected-class confidence is max(p1, 1 - p1).

- A predicts 1 with confidence 0.95 and is correct. F predicts 0 with confidence 0.90 and is wrong: p1 = 0.10 is not low confidence in the selected class.

  Reference labels are supplied for this exercise; real investigations must check their quality too.

- A score alone cannot establish reliability. **Calibration** asks whether, across many comparable cases, **prediction**s assigned a probability such as 0.90 are correct about 90 percent of the time.

  One correct or incorrect case cannot answer that question.

Transfer learning

> **My remembered idea, example and correction**

Embeddings and neighbourhoods

My remembered idea, example and correction

Training a small image model

My remembered idea, example and correction

Lesson reference for workbook practice

Case and image	p1	Reference label
A clear	0.95	1
B blurred	0.90	0
C clear	0.80	1
D blurred	0.60	0
E clear	0.45	0
F blurred	0.10	1

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main six cases (p1, reference): A (.95, 1), B (.90, 0), C (.80, 1), D (.60, 0), E (.45, 0), F (.10, 1). Predict 1 at p1 ≥ .50; selected confidence is max(p1, 1–p1). Predictions 1, 1, 1, 1, 0, 0; confidences .95, .90, .80, .60, .55, .90. Correct A/C/E: 3/6. TP 2, TN 1, FP B/D, FN F. Cutoff .80 accepts A/B/C/F; coverage 4/6, accepted accuracy 2/4; D/E deferred.

  Cutoff .95 accepts only A: coverage 1/6, accepted accuracy 1/1, five deferred. Cutoff .99 accepts none: coverage 0/6, accepted accuracy undefined. No reviewer outcomes supplied. Clear images 3/3 correct, blurred 0/3, a pattern without causal proof. Check labels and preprocessing; use new paired blur comparisons with related source versions kept in one split.

  Separate picture: A (.9, 0), B (.6, 1), C (.2, 0); confidence .9, .6, .8. Cutoff .8 accepts A/C, one correct of two, coverage 2/3. Cutoff .85 accepts wrong A, accuracy 0/1, coverage 1/3. Scores are supplied rather than measured model outputs; calibration and final system reliability are unknown.

- For B, E and F, give the predicted class, selected-class confidence and correctness. Explain why p1 = 0.10 does not mean F has low confidence in its predicted class.

My reasoning and evidence

W2 · Trace the calculation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main six cases (p1, reference): A (.95, 1), B (.90, 0), C (.80, 1), D (.60, 0), E (.45, 0), F (.10, 1). Predict 1 at p1 ≥ .50; selected confidence is max(p1, 1–p1). Predictions 1, 1, 1, 1, 0, 0; confidences .95, .90, .80, .60, .55, .90. Correct A/C/E: 3/6. TP 2, TN 1, FP B/D, FN F. Cutoff .80 accepts A/B/C/F; coverage 4/6, accepted accuracy 2/4; D/E deferred.

  Cutoff .95 accepts only A: coverage 1/6, accepted accuracy 1/1, five deferred. Cutoff .99 accepts none: coverage 0/6, accepted accuracy undefined. No reviewer outcomes supplied. Clear images 3/3 correct, blurred 0/3, a pattern without causal proof. Check labels and preprocessing; use new paired blur comparisons with related source versions kept in one split.

  Separate picture: A (.9, 0), B (.6, 1), C (.2, 0); confidence .9, .6, .8. Cutoff .8 accepts A/C, one correct of two, coverage 2/3. Cutoff .85 accepts wrong A, accuracy 0/1, coverage 1/3. Scores are supplied rather than measured model outputs; calibration and final system reliability are unknown.

- For all six cases, list false positives and false negatives, count true positives and true negatives, and calculate overall accuracy. Why might a single accuracy hide useful information?

My reasoning and evidence

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main six cases (p1, reference): A (.95, 1), B (.90, 0), C (.80, 1), D (.60, 0), E (.45, 0), F (.10, 1). Predict 1 at p1 ≥ .50; selected confidence is max(p1, 1–p1). Predictions 1, 1, 1, 1, 0, 0; confidences .95, .90, .80, .60, .55, .90. Correct A/C/E: 3/6. TP 2, TN 1, FP B/D, FN F. Cutoff .80 accepts A/B/C/F; coverage 4/6, accepted accuracy 2/4; D/E deferred.

  Cutoff .95 accepts only A: coverage 1/6, accepted accuracy 1/1, five deferred. Cutoff .99 accepts none: coverage 0/6, accepted accuracy undefined. No reviewer outcomes supplied. Clear images 3/3 correct, blurred 0/3, a pattern without causal proof. Check labels and preprocessing; use new paired blur comparisons with related source versions kept in one split.

  Separate picture: A (.9, 0), B (.6, 1), C (.2, 0); confidence .9, .6, .8. Cutoff .8 accepts A/C, one correct of two, coverage 2/3. Cutoff .85 accepts wrong A, accuracy 0/1, coverage 1/3. Scores are supplied rather than measured model outputs; calibration and final system reliability are unknown.

- At cutoff 0.80, list accepted and deferred cases, **coverage** and accepted-case accuracy. Can you calculate final accuracy after human review from the supplied data?

  What does the report function return for accepted-case accuracy at cutoff 0.99?

My reasoning and evidence

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main six cases (p1, reference): A (.95, 1), B (.90, 0), C (.80, 1), D (.60, 0), E (.45, 0), F (.10, 1). Predict 1 at p1 ≥ .50; selected confidence is max(p1, 1–p1). Predictions 1, 1, 1, 1, 0, 0; confidences .95, .90, .80, .60, .55, .90. Correct A/C/E: 3/6. TP 2, TN 1, FP B/D, FN F. Cutoff .80 accepts A/B/C/F; coverage 4/6, accepted accuracy 2/4; D/E deferred.

  Cutoff .95 accepts only A: coverage 1/6, accepted accuracy 1/1, five deferred. Cutoff .99 accepts none: coverage 0/6, accepted accuracy undefined. No reviewer outcomes supplied. Clear images 3/3 correct, blurred 0/3, a pattern without causal proof. Check labels and preprocessing; use new paired blur comparisons with related source versions kept in one split.

  Separate picture: A (.9, 0), B (.6, 1), C (.2, 0); confidence .9, .6, .8. Cutoff .8 accepts A/C, one correct of two, coverage 2/3. Cutoff .85 accepts wrong A, accuracy 0/1, coverage 1/3. Scores are supplied rather than measured model outputs; calibration and final system reliability are unknown.

- All three blurred cases are wrong. A team claims blur caused the errors and plans to put clear images in **training** and their blurred copies in final evaluation.

  Identify the unsupported claim and the split problem; propose a better investigation.

> **My reasoning and evidence**

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give **prediction**s and confidences, overall accuracy, and accepted cases at cutoff 0.80.

  Calculate **coverage** and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Analyse new outputs independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Add a labelled uncertainty and failure analysis **evidence** sheet to Image-pattern Network Investigation. Show how you can distinguish a predicted class, **confidence score** and observed correctness.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**
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MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Count both types of error for Uncertainty and failure analysis. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Uncertainty and failure analysis. Proposed explanation: "A low p1 always means low confidence". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Four new cases have (p1, reference label): K (0.85, 0), L (0.70, 1), M (0.30, 0), N (0.05, 0). Give **prediction**s and confidences, overall accuracy, and accepted cases at cutoff 0.80.

  Calculate **coverage** and accepted-case accuracy. Does this policy improve accuracy on the accepted subset here?

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

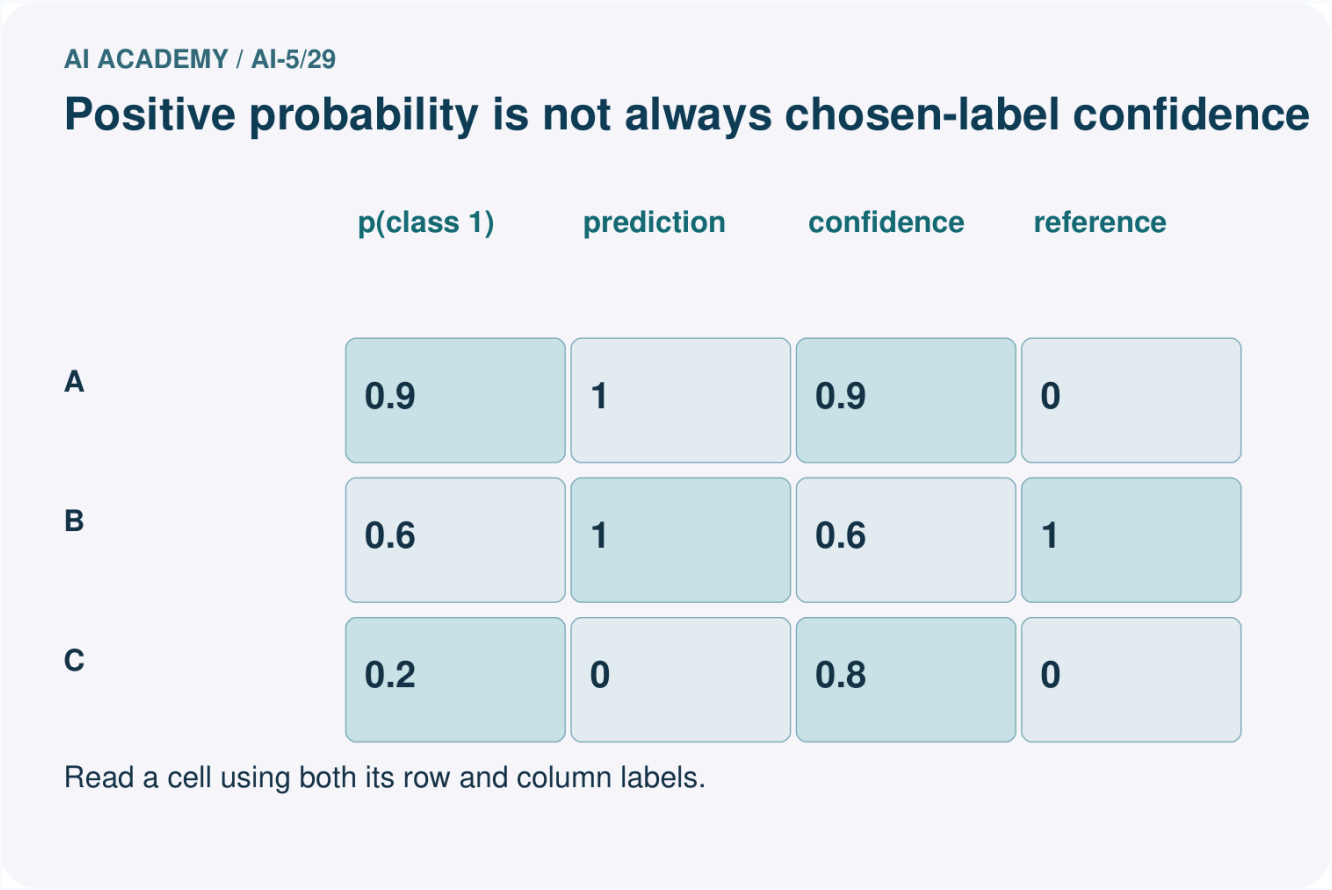
> Keep this task's inputs,rule and response together.

Complete task and evidence

- Keep positive-class probability separate from confidence in the chosen label.
- Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p).
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

- What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/29

Uncertainty and failure analysis

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Predictions

A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct.



2. Selected-label confidence

Confidences are 0.9, 0.6, 0.8. C's confidence is 0.8 despite p(class 1)=0.2.



3. Selective review

Accept confidence>=0.8: A and C; accuracy 1/2 and coverage 2/3.

CHANGE ONE THING

Raise the acceptance cutoff to 0.85.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project milestone

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.
- For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week’s input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ **Extra practice · Predict,check and repair**

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Keep positive-class probability separate from confidence in the chosen label.
- Which labelled step matches this detail? Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.
- 1. Selected-label confidence
- 2. Selective review
- 3. **Prediction**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Selective review Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Changed-case question and optional hint

- Changed-case practice: Raise the acceptance cutoff to 0.85. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. Only A is accepted: accuracy 0/1, **coverage** 1/3.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: Only A is accepted: accuracy 0/1, **coverage** 1/3. The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Repair and retain the record

Review Transfer learning (ai-5/28). Goal: Trace a reused feature extractor and a newly fitted **output** head. Short practice: Which part was trained for the new task?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Embeddings and neighbourhoods (ai-5/26). Goal: Look up a vector using a token ID and preserve sequence shape. Short practice: What would justify claiming nearby vectors mean related topics?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review **Training** a small image model (ai-5/23). Goal: Trace a complete image model with fixed features and one trained weight. Short practice: Does lower **training** loss alone establish that four updates generalise better?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

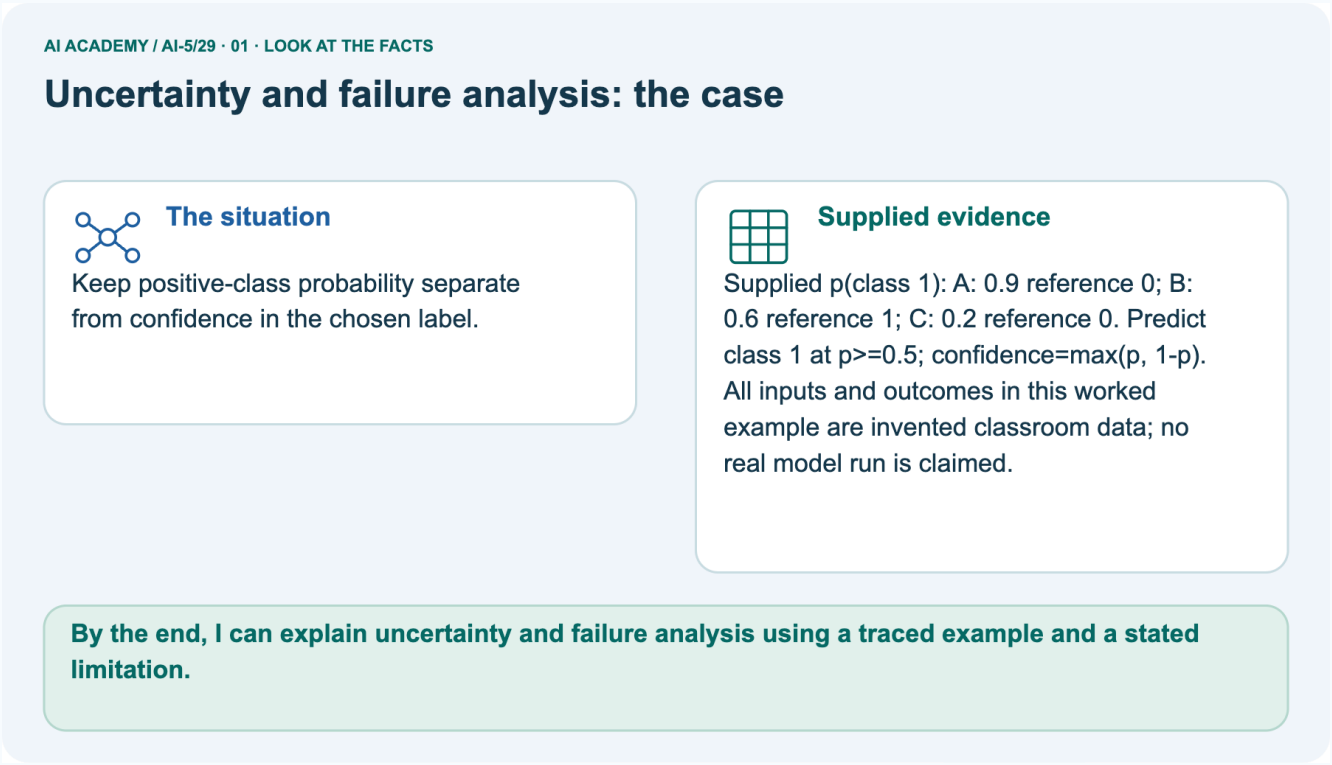
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Uncertainty and failure analysis: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

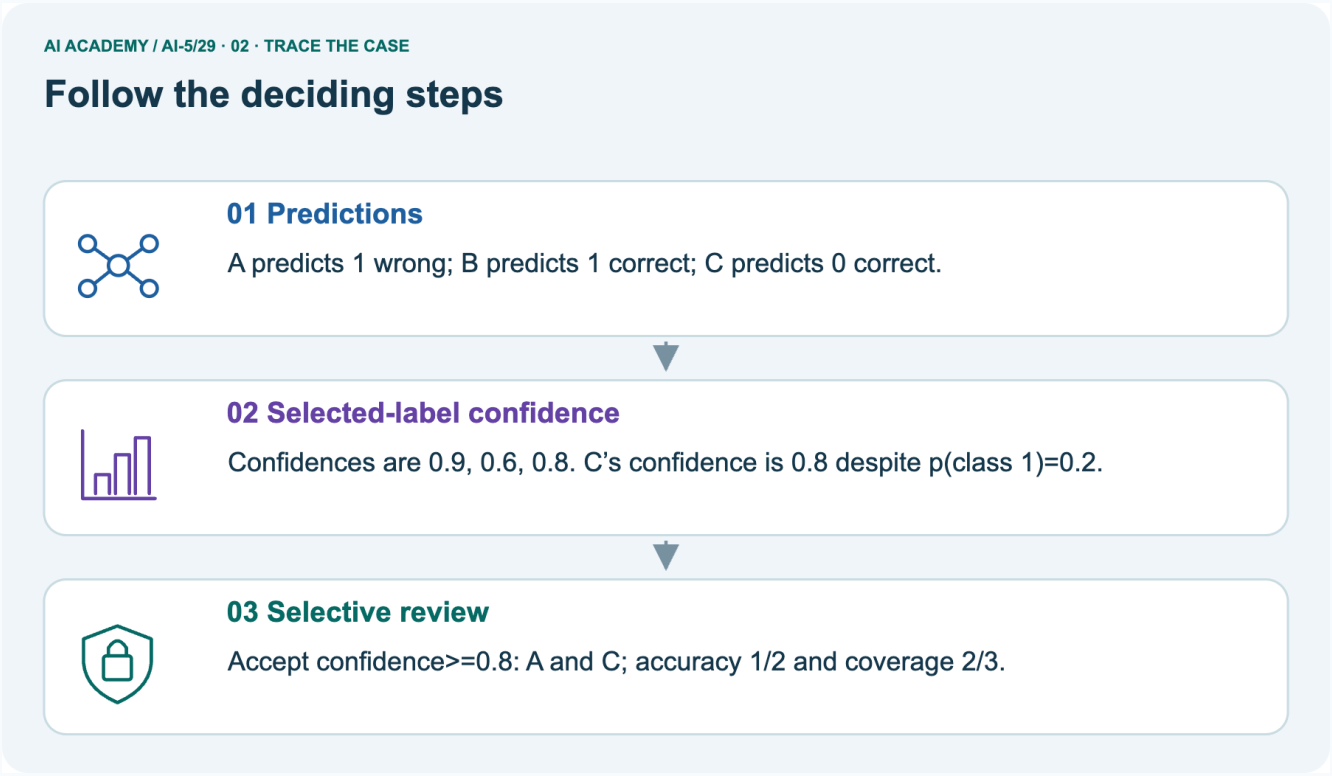
Keep positive-class probability separate from confidence in the chosen label. Supplied facts: Supplied p(class 1): A: 0.9 reference 0; B: 0.6 reference 1; C: 0.2 reference 0. Predict class 1 at p>=0.5; confidence=max(p, 1-p).

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

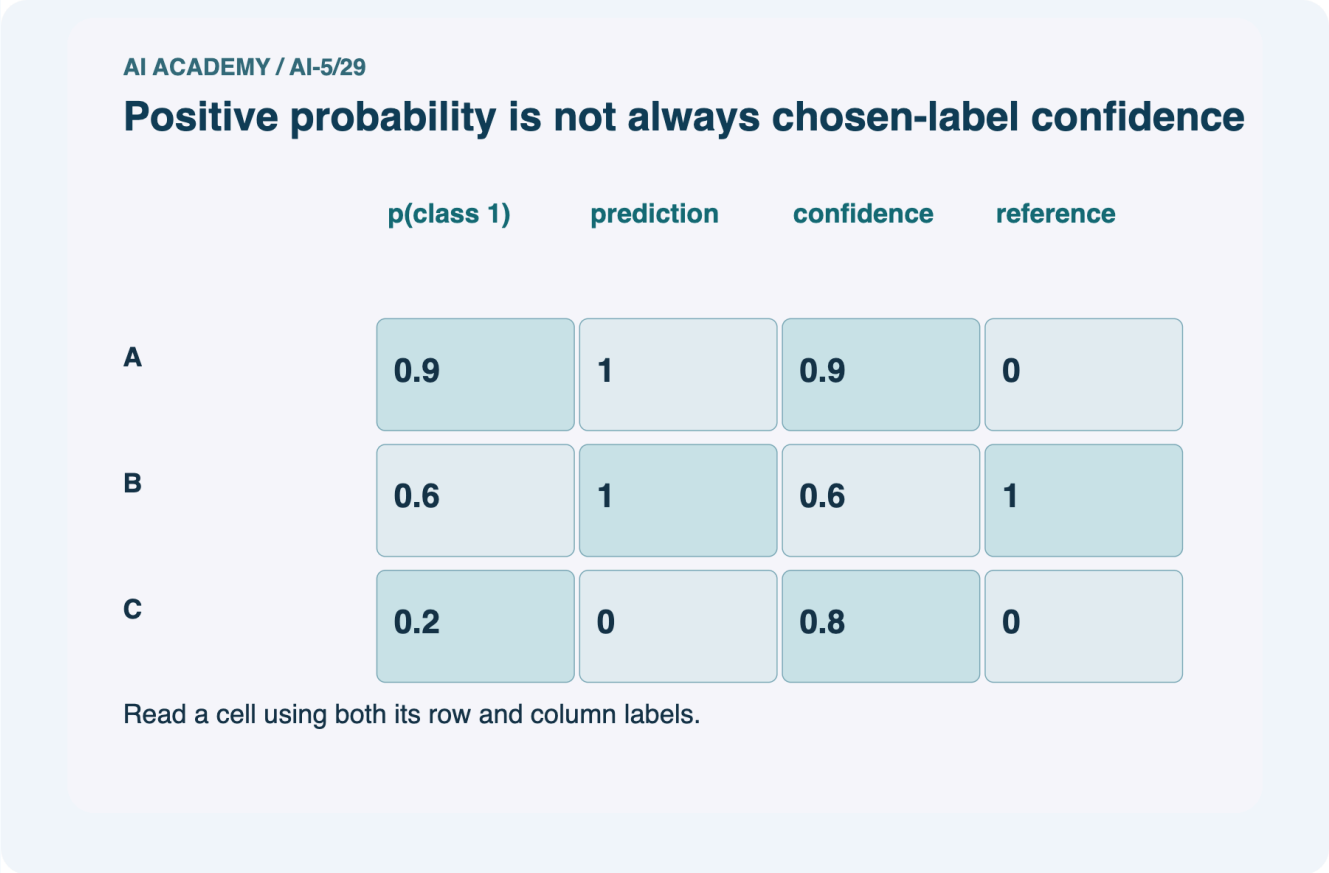
Predictions: A predicts 1 wrong; B predicts 1 correct; C predicts 0 correct. → Selected-label confidence: Confidences are 0.9, 0.6, 0.8.

C's confidence is 0.8 despite p(class 1)=0.2. → Selective review: Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Positive probability is not always chosen-label confidence



Follow the labelled picture with the matching explanation. Open the vector diagram

Positive probability is not always chosen-label confidence. A: p(class 1) = 0.9, **prediction** = 1, confidence = 0.9, reference = 0. B: p(class 1) = 0.6, **prediction** = 1, confidence = 0.6, reference = 1.

C: p(class 1) = 0.2, **prediction** = 0, confidence = 0.8, reference = 0.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.

AI ACADEMY / AI-5/29 · 04 · COMPARE THE CASES

Change one thing. Trace what follows.

Original case

Accept confidence>=0.8: A and C; accuracy 1/2 and coverage 2/3.

Changed condition

Raise the acceptance cutoff to 0.85.

Worked changed-case result

Only A is accepted: accuracy 0/1, coverage 1/3.
Why: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Follow the labelled picture with the matching explanation. Open the vector diagram

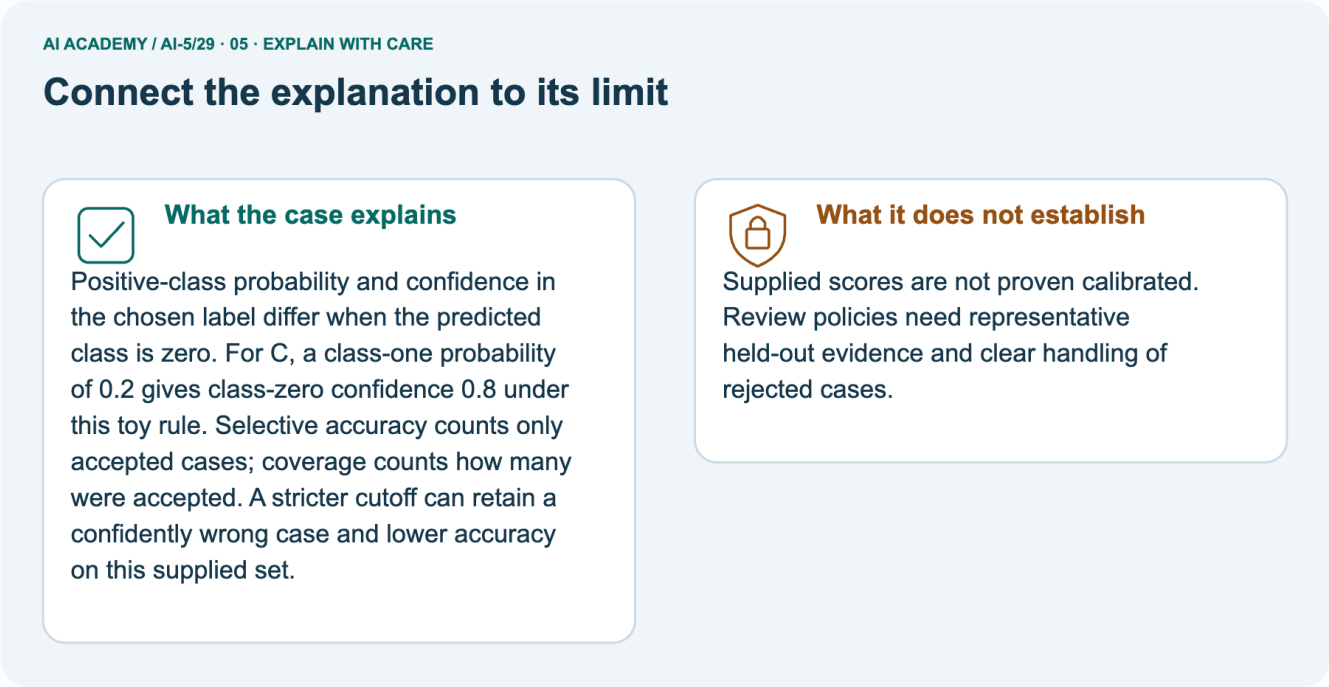
Original case: Accept confidence>=0.8: A and C; accuracy 1/2 and **coverage** 2/3. Changed condition: Raise the acceptance cutoff to 0.85. Expected result: Only A is accepted: accuracy 0/1, **coverage** 1/3.

Reason: The most confident supplied case is wrong; a stricter cutoff is not guaranteed to improve this set's accuracy.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Positive-class probability and confidence in the chosen label differ when the predicted class is zero. For C, a class-one probability of 0.2 gives class-zero confidence 0.8 under this toy rule.

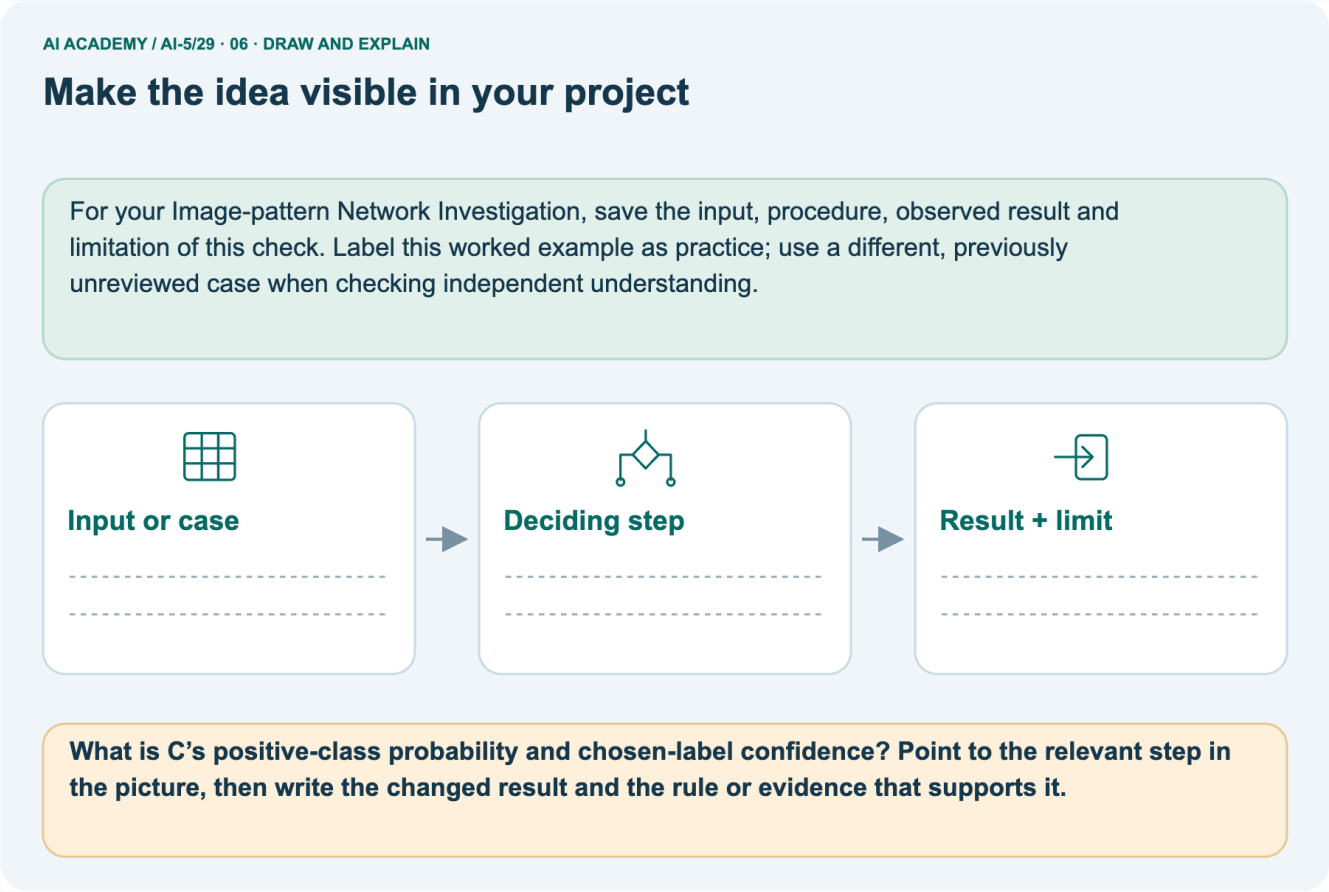
Selective accuracy counts only accepted cases; **coverage** counts how many were accepted. A stricter cutoff can retain a confidently wrong case and lower accuracy on this supplied set.

Limit: Supplied scores are not proven calibrated. Review policies need representative held-out **evidence** and clear handling of rejected cases.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What is C's positive-class probability and chosen-label confidence? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Uncertainty and failure analysis: the case

02. Follow the deciding steps

03. Positive probability is not always chosen-label confidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.